

SKILLS

- **AI / GenAI:** LLMs, RAG, LangChain, LangGraph, Agentic Workflows, MCP Gateway, Prompt Engineering, Tool/Function Calling, Embeddings, Vector DBs (pgvector, Pinecone), OpenAI API, HuggingFace, LLM Evaluation & Tracing
- **AI Systems & MLOps:** Orchestration, Guardrails, Human-in-the-Loop, Inference Optimization, OpenTelemetry, Distributed Tracing, Anomaly Detection, AIOps, Responsible AI, Model Risk
- **Languages:** Python, TypeScript, JavaScript, Go, SQL | Node.js, React, REST APIs
- **Cloud & Infra:** AWS (Bedrock, SageMaker, Lambda, EKS, Kinesis, S3), GCP Vertex AI, Kubernetes, Terraform, Docker, CI/CD
- **Data & Platforms:** Kafka, Airflow, Snowflake, Elasticsearch, Redis, ETL pipelines

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – Applied AI Nov 2022 – Present
 - **ASB Assist – Agentic GenAI Platform (In Development):** Building conversational AI agents on RBC’s enterprise blueprint platform: natural-language requirements in, compliant architecture diagrams and service recommendations out; LLM + RAG pipeline with LangChain orchestration, MCP gateway and agent-gateway tool integration, prompt/context engineering, and iterative offline evaluation of answer quality and safety.
 - **Enterprise AI Governance & Responsible AI:** Defined and operationalised org-wide AI governance covering Responsible AI, model risk, IAM, audit trails, and regulatory compliance (OSFI, GDPR) across AWS, GCP, and OCP; aligned guardrails and approval workflows for production AI features in a regulated financial institution.
 - **AI/ML Observability & Data Platform (LAS):** Architected institution-wide telemetry (Kafka, OpenTelemetry, Vector, Logstash) ingesting 150 TB/day across 10,000+ systems; standardised schemas and collection pipelines enabling AIOps, model monitoring, drift detection, and latency SLOs for distributed AI services; OTel guides adopted by 20+ teams.
 - **0-to-1 Platform Delivery (ASB Builder, Patent-Pending):** Led production delivery of enterprise blueprint platform (Node.js/TypeScript, React) with rules-based recommendation engine and auto-certification—foundation layer for ASB Assist; org-wide adoption in air-gapped environments; wrote 99.6% of production code and 100% of core logic.
 - **Team Leadership & Cross-Functional Delivery:** Led 5 engineers; partnered with product, risk, security, and architecture on technical design, ADRs, code reviews, and sprint delivery; mentored engineers on cloud and applied AI practices.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Production ML Inference Pipeline:** Deployed transformer-based document classification on AWS EKS with inference APIs, monitoring, and HIPAA audit logging; reduced manual review by 60%; end-to-end ownership from model integration through production operations.
 - **0-to-1 Full Stack Platform:** Built React/TypeScript frontend, Node.js APIs, and verification layer from scratch as first engineer; scaled team to 20+ before acquisition.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op Apr 2022 – Nov 2022
 - **Platform Modernisation:** Migrated federal tax platform UI to React/TypeScript; contributed to digital services APIs serving millions of users with accessibility and quality gates.

PROJECTS

- **Enterprise Document Q&A with Eval Harness (Personal):** LangChain + pgvector RAG over document corpora with offline eval sets, regression metrics, and prompt/version comparison—patterned after production LLM evaluation workflows for safety and quality iteration.
- **Flight Price Optimization Agent:** DQN/DDQN reinforcement-learning agent (Python); historical flight data; 18% improvement over baseline in simulation; SageMaker/Lambda inference prototype.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020